

Tailoring Machine Learning Weather Predictions for Local Impacts

Ozma Houck, James Franke

Advancements in ML based Weather Forecasting have made Tailored Forecasting Models Possible

- Traditional numerical weather prediction (NWP) models are expensive and slow to run.
- ML based weather models are fast to run, but are still expensive to train.
- These models predict the entire globe at once. A farmer in cares a lot about if it will rain in five days. Less about wind speeds over in the troposphere over Greenland.

Advancements in ML based Weather Forecasting have made Tailored Forecasting Models Possible

- Traditional numerical weather prediction (NWP) models are expensive and slow to run.
- ML based weather models are fast to run, but are still expensive to train.
- These models predict the entire globe at once. A farmer in cares a lot about if it will rain in five days. Less about wind speeds over in the troposphere over Greenland.

Goal: Demonstrate that is is possible to meaningfully improve the accuracy of relevant weather variables using a small amount of compute power and data.

Advancements in ML based Weather Forecasting have made Tailored Forecasting Models Possible

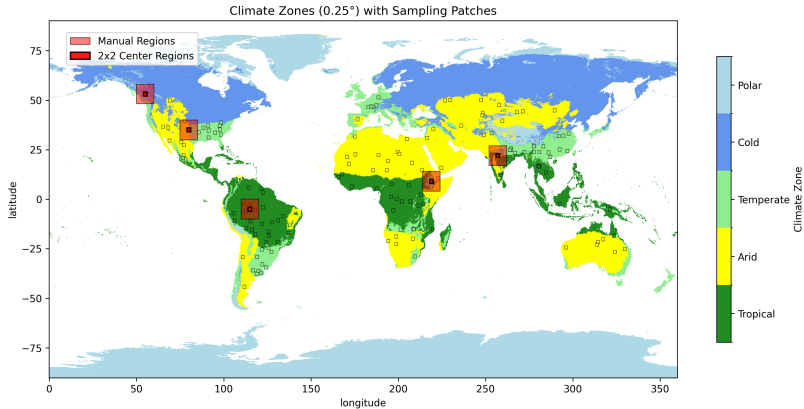
- Traditional numerical weather prediction (NWP) models are expensive and slow to run.
- ML based weather models are fast to run, but are still expensive to train.
- These models predict the entire globe at once. A farmer in cares a lot about if it will rain in five days. Less about wind speeds over in the troposphere over Greenland.

Goal: Demonstrate that is is possible to meaningfully improve the accuracy of relevant weather variables using a small amount of compute power and data.

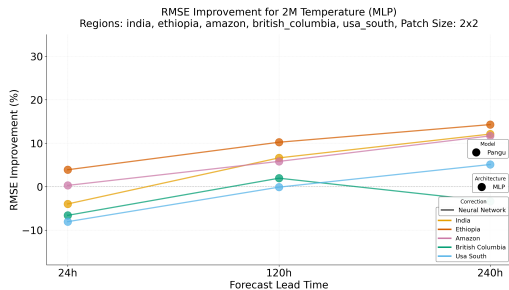
Contribution: Apply neural network post-processing model to both traditional and ML weather models, showing that it is possible to improve the accuracy of both (perhaps with some caveats).

Start by Manually Looking at Five Regions of Interest

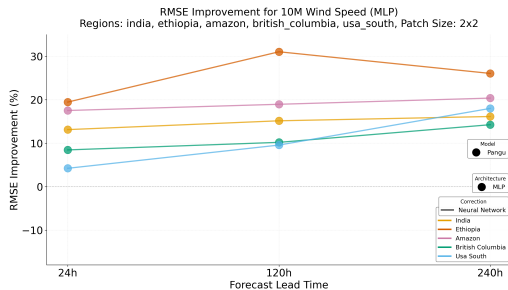
- Manually choose a few regions with a variety of climates and land terrains.
- Sample from different climate zones.



See Greater Improvements for Longer Lead Times and for Wind Speed

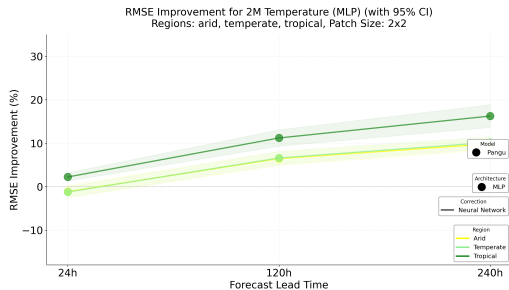


(a) 2m Temperature: Pangu (ML Model)

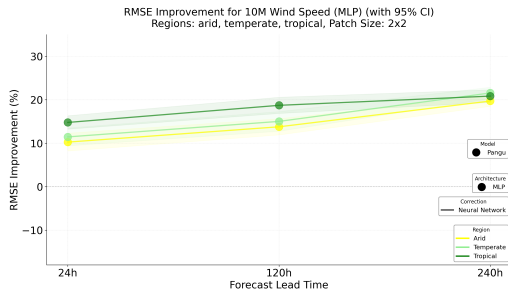


(b) Wind Speed: Pangu (ML Model)

Climate Zones



(a) 2m Temperature: Pangu



(b) Wind Speed: Pangu